EFS03 & 04: Objectives and Prerequisites

Key Takeaways from the lecture:

The module focuses on the theories and practical implementation of momentum strategies using Python. Basic knowledge of Python is assumed, and some conceptual programming language would be helpful. The math requirement assumed is basic college-level statistics.

Course outline:

1. **Causes of momentum**
    a. Persistence of futures roll returns.
    b. Slow diffusion of news.
    c. Forced sales and purchases by funds.
    d. HFT market manipulation.
2. **Roll returns as driver of momentum**
    a. Backwardation vs. contango.
        - Exercise: Estimating spot and roll returns.
    b. Time-series vs cross-sectional momentum.
    c. Arbitrage between future and spot returns.
        - The case of VX-ES.
    d. Statistical tests for time-series momentum.
    e. Example futures time-series momentum strategy.
        - Indicators for TS momentum.
    f. Example futures cross-sectional momentum strategy.
    g. Example stock cross-sectional momentum strategy.
        - Indicators for CS momentum.
        - News sentiment.
    h. The phenomenon of “Momentum Crashes”.
        - The S&P DTI index.

3. **Forced sales and purchases due to funds**
    a. Hedge funds.
    b. Mutual funds.
        - Example strategy using Pressure indicator.
    c. Index funds.
    d. Levered ETFs.
        - Example strategy.
4. **Exit Strategies**
5. **Advantages and disadvantages of momentum strategies as compared to mean reversion strategies.**

Prerequisites before the lecture:

- EFS-01 Session.
- EFS-02 Session.
- The Python module can be completed.

Download files to be used in the lecture:

- Please note that all files required for the lecture have been uploaded to the LMS under the section 'EFS-03 & 04: Quantitative Momentum Strategies – I & II